

Curriculum Vitae (CVI)

Miguel A. Mendez

Personal Details

Family name: Mendez
First name: Miguel Alfonso
Date and place of birth: 31/10/1987, Maracay (Venezuela)
Nationality: Italian
ORCID: [0000-0002-1115-2187](https://orcid.org/0000-0002-1115-2187)
Website: mendezma.com
Google Scholar: [Scholar profile](#)
ResearchGate: [RG Profile](#)



Current Position

- **Associate Professor**, von Karman Institute for Fluid Dynamics (VKI), Environmental & Applied Fluid Dynamics Department, Sint-Genesius-Rode, Belgium (May 2023–present).
- **Honorary Lecturer (interim appointment)**, Aero-Thermo-Mechanics Laboratory, École Polytechnique de Bruxelles, Université Libre de Bruxelles, Brussels, Belgium (February 2024–present)
- **Profesor Honorífico**, Experimental Aerodynamics and Propulsion Lab, Universidad Carlos III de Madrid, Leganés, Spain (February 2025–present).

Education

- **PhD in Engineering**, Université libre de Bruxelles (2013–2018). Thesis: *Dynamics of Gas Jet Impinging on Falling Liquid Films*. Supervisor: Jean Marie Buchlin (VKI).
- **Research Master in Experimental Fluid Mechanics**, von Karman Institute (2012–2013). Graduated with honours; von Karman Prize.
- **MSc in Energy and Nuclear Engineering**, University of Florence (2009–2012). GPA 110/110 *cum laude*.
- **5th Year In Industrial Engineering**, ETSI Sevilla (2010–2011). GPA 8.9/10.
- **BSc in Mechanical Engineering**, University of L'Aquila (2006–2009). GPA 107/110.

Previous Positions

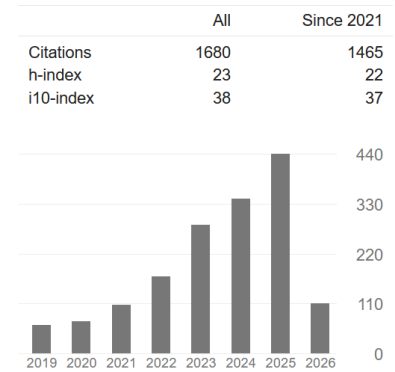
- **Assistant Professor**, VKI, Environmental & Applied Fluid Dynamics Department (2019–2023).
- **Postdoctoral Researcher**, VKI, Environmental & Applied Fluid Dynamics Department (2018–2019).

Awards & Distinctions

Emerging Leader 2022, by *Measurement Science and Technology*. **Outstanding Paper 2020**, by *Measurement Science and Technology*. **Solvay PhD Thesis Award** for Research Excellence (2019). **AFVL PhD Thesis Award** for laser diagnostics in fluid mechanics, by AFVL. **Best Oral Presentation at the Annual VKI PhD Symposium 2015-2016**. **Best Oral Presentation at the 10th Experimental Fluid Mechanics Conference 2015** **Theodore von Karman Prize** (best performance at the VKI Research Master program)

Research Output and Funding

- **Research Output & Impact** ≈ 70 journal papers, 90+ conferences, 10 chapters in books; 2 edited books. The evolution of citations from Google Scholar is shown on the right. RG score: 2,929 Research Interest on Research Gate (top 3%); Google scholar h-index: 23; i10-index: 38.
- **Funding (last 5 years):** ~ 3.1 M€ as PI (including 1.5 M€ for an ERC Starting Grant) and ~ 3.4 M€ as collaborator on specific project work packages (see 6_PIN document).



Teaching (A.Y. 2025/2026)

(with starting year of each teaching activity)

- Fundamentals of Fluid Dynamics (22 h, 2.5 ECTS), VKI. *Started in 2017 as DEFM, adapted in 2019*
- Signal Processing, Module II (12 h, 1.5 ECTS), VKI. *Started in 2019*
- Introduction to Measurement Techniques (25% of the course), VKI. *Started in 2017*
- Tools for Scientific Computing (12 h, 1 ECTS), VKI *Started in 2022*
- Data-Driven Modal Analysis (12 h, 1 ECTS), VKI *Started in 2022*
- Machine Learning for Fluid Dynamics, Module I (12 h, 1 ECTS) at VKI. *Started as Data Driven Fluid Mechanics (DDFM) in 2021, adapted in 2025.*
- Machine Learning for Fluid Dynamics, Module II (24 h, 2 ECTS), VKI. *Started as DDFM since 2021, adapted in 2025.*
- Digital Twinning, Data Assimilation and Control (18 h, 2 ECTS), VKI. *Started in 2024.*
- Hands-on Machine Learning for Fluid Dynamics (32 h, 3 ECTS), VKI. *Started in 2021.*
- *Data-Driven Engineering (MECAH419)* (10 h), ULB. *Started in 2022*

Service to the Profession and Society

- Reviewer for: *Data-Centric Engineering, Journal of Fluid Mechanics, Physics of Fluids, Experimental Thermal and Fluid Science, Review of Scientific Instruments, Chemical Engineering Science, International Journal of Multiphase Flows, Industrial & Engineering Chemistry Research, Journal of Coatings Technology and Research, European Journal of Operational Research.*
- Organizer/director and main lecturer: *Hands-on Machine Learning for Fluid Dynamics* 2021-2026 (yearly), >650 participants so far.
- Co-organizer/director: VKI Lecture Series on PIV/LPT (2021, 2024); *Introduction to Measurement Techniques* (2020); *Data Driven Fluid Mechanics* LS (2020, >230 participants; lecture notes published by Cambridge University Press), *Machine Learning for Fluid Dynamics* LS (2024, >300 participants; lecture notes published by von Karman Institute) ; *Flow Control LS* (2026)
- Scientific Committee member: *European Coating Symposium, AI Fluids, International Symposium on Flow Visualization (ISFV).*

Supervision and PhD Committee Service

- I currently supervise 11 PhD students (8 as PI and 3 in international co-supervision), with five defenses completed in 2023–2024.
 - Visiting PhDs (2023-2025): L. Kyriakidis, (DLR), T. Werner (TU Darmstadt), D. Gligor (UPM), I. Tirelli (UC3M), E. Saccaggi (Polito), J.Wang (Monash Univ), D. Grasev (UoD), M. Belda (Cz. Techn Univ)
 - I currently supervise 4 Research Master students at VKI (26 supervised in the last 5 years). 100+ MSc theses/internships supervised (full list in 6_PES).
 - Member of 11 PhD Committees: Kamila Zdybal (ULB), Mushin Akkurt (VKI-UGhent), Sagar Adatreo (TU-DELFT), Antonio Colanera (UniNapoli), Hang Wang (ULB), Michele Quattromini (Univ Paris Saclay), Ivan Kharsansky (Ensta Paris), Enrico Amico (PoliTO), Riccardo Margheriti (PoliMi), Lorenzo Giunti (UniFI), Adrian Grille Guerra (TU Delft).
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Outreach and Engagement

- **Public outreach:** Contributor to the VKI Open Days, presenting fluid-dynamics demonstrations and research activities to the general public.
 - **Science communication:** Featured in a popular podcast on ‘Engineering & PhD Life,’ discussing fluid-dynamics research, PhD training, and career perspectives ([link](#)), and in a recorded panel organized by NCP-FNRS to share feedback and strategic advice for prospective European-grant applicants (ERC, Horizon Europe) ([link](#)).
 - **Open-source software and training:** Developer of the open-source tools MODULO (MODal mUltiscale pOd, see [github](#)) for modal decomposition and data-driven reduced-order modelling and SPICY (meshless Pressure from Image velocimetry, see [github](#)) for physics-constrained radial-basis-function regression for scattered data. Both tools are accompanied by public tutorials and instructional videos for the research community (see [MODULO on youtube](#), see [SPICY on youtube](#)).
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Name	University	Co-supervisor	Funding	Year	2023	2024	2025	2026	2027	2028
Domenico Fiorini	KULeuven	D. Seveno	FWO/ESA	Defended						
Fabio Pino	ULB	B. Scheid	FRIA/AM	Defended						
Matilde Fiore	UCLouvain	Y. Bartosiewicz	FRIA/SCK	Defended						
David Barreiro	UCoruna	A. Gosset	UCoruna/AM	Defended						
Romain Poletti	UGhent	J. Degroote	FWO	Defended						
Pedro Marques	ULB	B. Scheid	FRIA/ESA	Defended						
Jan van Den Berghe	UCLouvain	Y. Bartosiewicz	FRIA	Defended						
Lorenzo Schena	VUB	J.Helsen	FWO	4rth						
Manuel Ratz	ULB	A. Parente	FRIA/ERC	3rd						
Damien Rigutto	ULB	B. Scheid	FRIA/AM	3rd						
Umberto Zucchelli	ISAE	F. Sanfedino	ESA	3rd						
Francisco Monteiro	UC3M	–	VLAIO/ERC	2nd						
Samuel Ahizi	UC3M	–	VLAIO/ERC	2nd						
Sebastiano Randino	ULB	E. Garone	FRIA/ERC	2nd						
David Grasev	UoD	J. Hnidka	ERC/UoD	2nd						
Tommaso De Maria	ULB	E. Garone	ERC	1st						
Yannick Lecomte	ULB	E. Garone	ERC	1st						

Table 1: Summary of PhD students and project timeline (gray = active years; light blue = current academic year). The team also includes three postdoctoral researchers: R. Poletti, D. Barreiro, and A. Grava.

Selected Invited Lectures, Seminars and Keynotes (2022–2025)

1. **Lectures at the EA2 “Advanced Post Processing Techniques Summer School”: AI in Fluid Dynamics — Meshless Modal Analysis of Scattered Data**, invited by Prof. Avallone, [event page](#), Torino, July 2025.
2. **Lectures at UC3M Doctoral School “An Introduction to Statistical and Variational Methods for Assimilation and Real Time Modeling”**, invited by Prof. S. Discetti, Madrid, Jun 2025, [link](#).
3. **Keynote Lecture on Reinforcement Twinning at the 1st AI and Fluid Mechanics Conference**, invited by Prof. Gavaises, [programme](#), Chania, May 2025.
4. **Seminar on Machine Learning for Fluid Dynamics at BrIAS**, invited by Prof. E. Garone, [video](#), Brussels, February 2025.
5. **Seminar on Reinforcement Twinning at the WP3 Cypher Meeting**, invited by Prof. A. Parente, [event page](#), Madrid, February 2025.
6. **Seminar on Meshless and Binless Super-resolution for PTV**, Exp in Fluids Seminar, invited by Prof. Tropea, [video](#), Online, November 2024.
7. **Seminar on Engineering AI at the workshop “Bridging Knowledge: Artificial Intelligence”**, organized by ASI, invited by Dr. A. Turchi, [agenda](#), Rome, May 2024.
8. **Seminar on “Trends and Challenges in Scientific Machine Learning for Engineering”**, invited by Dr. Esposito, ASI Online Seminar Series, [event page](#), March 2024.
9. **Keynote Lecture on Reinforcement Twinning at the 1st Cypher General Meeting**, invited by Prof. A. Parente, [meeting page](#), Ljubljana, April 2024.
10. **Keynote Lecture on Machine Learning for Fluid Dynamics at the 12th AIAC**, invited by Prof. Özgen, METU, Ankara, September 2023, [conference site](#).
11. **Lecture on Machine Learning for Fluids at the Short Course on CFD of ICCS**, invited by Dr. Shane, London, September 2023, [course programme](#).
12. **Three Lectures on Deep Learning for Turbulence Modeling at the FJOH Summer School**, invited by Prof. R. Stieglitz, Karlsruhe, August 2023, [school page](#), [video](#).
13. **Seminar at Extrality: “Trends and Challenges in Scientific Machine Learning for Fluid Dynamics”**, invited by Dr. Mazari, Online, September 2023, [event link](#).
14. **Seminar on Machine Learning for Engineering at IMT Nord Europe**, invited by Prof. Lacassagne, Douai, June 2023.
15. **Five Lectures on Data-Driven Modal Analysis at CAMS–AUB**, invited by Prof. Najem, Beirut, February 2023, [programme](#), [video](#).
16. **Lecture at JT51 on Data Processing**, invited by Prof. Zimmer, Paris, November 2022, [event page](#).
17. **Seminar at City, University of London: “Data-Driven Fluid Mechanics”**, invited by Prof. Marinos Manoles, London, October 2022, [event page](#).
18. **Seminar at KTH: “Comparative Analysis of Machine Learning Methods for Active Flow Control”**, invited by Prof. R. Vinuesa, Stockholm, September 2022, [event page](#).
19. **Seminar at Universidade da Coruña (UDC) on Machine Learning for Fluid Dynamics**, invited by Prof. Gosset, Ferrol, September 2022, [event page](#).
20. **Keynote Lecture on Data-Driven Methods for Image Velocimetry at the CFTL 2022 Meeting**, invited by Prof. Vetrano, Leuven, September 2022, [meeting page](#).
21. **Lecture on Model-Free Control Methods at the Workshop on Complex Fluids, Centro Enrico Fermi**, invited by Prof. Biferale, Rome, July 2022, [programme](#).

List of Publications

Under Review

- [1] D. Barreiro-Villaverde, A. Cantiani, and M. A. Mendez. “On the complex interplay of temperature, phase change and natural convection in self-pressurization-an investigation using segregated modeling”. Submitted to International Journal of Heat and Mass Transfer. 2025. arXiv: [2512.13349](#).
- [2] D. Rigutto, M. Ratz, and M. A. Mendez. “A meshless data-tailored approach to compute statistics from scattered data with adaptive radial basis functions”. Submitted to Experiments in Fluids. 2025. arXiv: [2511.20449](#).
- [3] F. Monteiro, T. De Maria, S. Ahizi, R. Abarca, G. C. A. Caridi, and M. A. Mendez. “Regime maps for sloshing in horizontal cylindrical tanks under vertical acceleration”. Submitted to Physical Review Fluids. 2025. arXiv: [2512.02540](#).
- [4] P. A. Marques, A. Simonini, B. Scheid, and M. A. Mendez. “Sloshing-induced pressure drop in cryogenic propellant tanks: an experimental analysis with flow visualization”. Submitted to International Journal of Heat and Mass Transfer. 2025.
- [5] M. Ratz, A. Parente, and M. A. Mendez. “Meshless data-driven decompositions with RBF-based inner products”. Submitted to Computers and Fluids. 2025. arXiv: [2511.03264](#).
- [6] S. Ahizi, F. Monteiro, R. Abarca, and M. A. Mendez. “Heat-transfer enhancement by parametric sloshing in horizontal cylinders: experiments and EKF-based identification of Nusselt numbers”. Submitted to International Journal of Heat and Mass Transfer. 2025. arXiv: [2510.19540](#).
- [7] J. Van den Berghe, M. A. Mendez, and Y. Bartosiewicz. “On the choking mechanism in supersonic ejectors: a one-dimensional analysis of Reynolds-averaged Navier–Stokes simulations”. Submitted to Journal of Fluid Mechanics. 2025. arXiv: [2510.23385](#).
- [8] S. Randino, L. Schena, N. Coudou, E. Garone, and M. A. Mendez. “Nonlinear system identification for model-based control of waked wind turbines”. Submitted to Data-Centric Engineering. 2025. arXiv: [2510.07336](#).
- [9] R. Poletti, E. Bombardi, L. Koloszar, M. A. Mendez, and J. Degroote. “Characterisation and extension of a rigid-body dynamics solver coupled with OpenFOAM for flight performance analysis of flapping-wing drones”. Submitted to OpenFOAM Journal. 2025. arXiv: [2510.24518](#).
- [10] A. K. N. Doan, K. Bizon, P. Cinnella, M. A. Mendez, and A. Parente. “Hybrid physics-based machine learning methods for renewable sustainable fuels modelling”. Roadmap article submitted to Journal of Physics: Energy (IOP). 2025.
- [11] F. Pino, E. Fracchia, B. Scheid, and M. A. Mendez. “Integral modelling and reinforcement-learning control of 3D liquid metal coating on a moving substrate”. Submitted to Physical Review Fluids. 2025. arXiv: [2503.14270](#).
- [12] R. Poletti, L. Schena, L. Koloszar, J. Degroote, and M. A. Mendez. “Reinforcement Twinning for hybrid control of flapping-wing drones”. Submitted to Data-Centric Engineering. 2025. arXiv: [2505.18201](#).

Journal Articles Published

- [13] Lorenzo Schena, Wim Munters, Jan Helsen, and Miguel A. Mendez. “POD-Based Sparse Stochastic Estimation of Dynamic Wind Turbine Blade Deflections”. In: *Journal of Sound and Vibration* (Mar. 2026), p. 119738. ISSN: 0022-460X. DOI: [10.1016/j.jsv.2026.119738](#).
- [14] Iacopo Tirelli, Miguel Alfonso Mendez, Andrea Ianiro, and Stefano Discetti. “Meshless super-resolution of scattered data via constrained radial basis functions and K-nearest-neighbors-driven densification”. In: *Physical Review Fluids* 11.2 (Feb. 2026). ISSN: 2469-990X. DOI: [10.1103/nv1c-1cmn](#).

- [15] P. A. Marques, S. Ahizi, F. Monteiro, B. Scheid, and M. A. Mendez. “On the scaling of heat and mass transfer in cryogenic propellant tanks: A model-based and experimental analysis under static conditions and lateral sloshing”. In: *Applied Thermal Engineering* 279 (2025). DOI: [10.1016/j.applthermaleng.2025.127751](https://doi.org/10.1016/j.applthermaleng.2025.127751).
- [16] L. Kyriakidis, M. Bähr, and M. A. Mendez. “Enhanced hybrid algorithm based on Bayesian optimization and Interior Point OPTimizer for constrained optimization”. In: *Optimization and Engineering* (2025). DOI: [10.1007/s11081-025-09975-y](https://doi.org/10.1007/s11081-025-09975-y).
- [17] I. Tirelli, M. A. Mendez, A. Ianiro, and S. Discetti. “A meshless method to compute the proper orthogonal decomposition and its variants from scattered data”. In: *Proceedings of the Royal Society A* (2025). DOI: [10.1098/rspa.2024.0526](https://doi.org/10.1098/rspa.2024.0526). arXiv: [2407.03173](https://arxiv.org/abs/2407.03173).
- [18] M. Fiore, E. Saccaggi, L. Koloszar, Y. Bartosiewicz, and M. A. Mendez. “Data-driven turbulent heat flux modeling with inputs of multiple fidelity”. In: *Physical Review Fluids* 10.3 (2025), p. 034606. DOI: [10.1103/PhysRevFluids.10.034606](https://doi.org/10.1103/PhysRevFluids.10.034606). arXiv: [2409.03395](https://arxiv.org/abs/2409.03395).
- [19] F. Monteiro, P. Marques, A. Simonini, L. Carbonelle, and M. A. Mendez. “Experimental characterization of non-isothermal sloshing in microgravity”. In: *Experimental Thermal and Fluid Science* 166 (2025), p. 111476. DOI: [10.1016/j.expthermflusci.2025.111473](https://doi.org/10.1016/j.expthermflusci.2025.111473). arXiv: [2410.06590](https://arxiv.org/abs/2410.06590).
- [20] F. Pino, M. A. Mendez, and B. Scheid. “Pareto front of the magnetic wiping process in dip coating”. In: *Journal of Engineering Mathematics* (2025). Issue 1/2025. DOI: [10.1007/s10665-025-10426-x](https://doi.org/10.1007/s10665-025-10426-x). arXiv: [2406.14110](https://arxiv.org/abs/2406.14110).
- [21] J. Van den Berghe, M. A. Mendez, and Y. Bartosiewicz. “An extension of the compound flow theory with friction between the streams and at the wall”. In: *Journal of Fluid Mechanics* 1000 (2024), A85. DOI: [10.1017/jfm.2024.1041](https://doi.org/10.1017/jfm.2024.1041). arXiv: [2401.07236](https://arxiv.org/abs/2401.07236).
- [22] F. Pino, M. A. Mendez, and B. Scheid. “Linear stability analysis of a vertical liquid film over a moving substrate”. In: *Journal of Fluid Mechanics* 1000 (2024), A57. DOI: [10.1017/jfm.2024.940](https://doi.org/10.1017/jfm.2024.940).
- [23] D. Barreiro-Villaverde, A. Gosset, and M. A. Mendez. “On the coupling instability of a gas jet impinging on a liquid film”. In: *Journal of Fluid Mechanics* 992 (2024), A11. DOI: [10.1017/jfm.2024.553](https://doi.org/10.1017/jfm.2024.553). arXiv: [2309.15502](https://arxiv.org/abs/2309.15502).
- [24] F. Pino, M. A. Mendez, and B. Scheid. “Absolute and convective instabilities in a liquid film over a substrate moving against gravity”. In: *Physical Review Fluids* 9 (2024), p. 104002. DOI: [10.1103/PhysRevFluids.9.104002](https://doi.org/10.1103/PhysRevFluids.9.104002). arXiv: [2312.14613](https://arxiv.org/abs/2312.14613).
- [25] L. Kyriakidis, M. A. Mendez, and M. Bahr. “A hybrid algorithm based on Bayesian Optimization and Interior Point OPTimizer for optimal operation of energy conversion systems”. In: *Energy* 312 (2024), p. 133416. DOI: [10.1016/j.energy.2024.133416](https://doi.org/10.1016/j.energy.2024.133416).
- [26] D. Gligor, P. A. Marques, P. S. Sanchez, J. Porter, M. A. Mendez, and J. M. Ezquerro. “Experiments on sloshing mitigation using tuned oscillating baffles”. In: *Physics of Fluids* 36 (2024), p. 092122. DOI: [10.1063/5.0225917](https://doi.org/10.1063/5.0225917).
- [27] D. Fiorini, A. Simonini, J. Steelant, D. Seveno, and M. A. Mendez. “An experimental characterization of capillary driven flows in microgravity”. In: *Microgravity Science and Technology* 36 (2024), p. 61. DOI: [10.1007/s12217-024-10142-8](https://doi.org/10.1007/s12217-024-10142-8).
- [28] R. Poletti, L. Schena, D. Ninni, and M. A. Mendez. “MODULO: a python toolbox for data-driven modal decomposition”. In: *Journal of Open Source Software* 9.102 (2024), p. 6753. DOI: [10.21105/joss.06753](https://doi.org/10.21105/joss.06753).
- [29] R. Poletti, A. Calado, L. Koloszar, J. Degroote, and M. A. Mendez. “On the unsteady aerodynamics of flapping wings under dynamic hovering kinematics”. In: *Physics of Fluids* 36 (2024), p. 081901. DOI: [10.1063/5.0215531](https://doi.org/10.1063/5.0215531). arXiv: [2408.03222](https://arxiv.org/abs/2408.03222).

- [30] M. Ratz and M. A. Mendez. “A meshless and binless approach to compute statistics in 3D ensemble PTV”. In: *Experiments in Fluids* 65 (2024), p. 142. DOI: [10.1007/s00348-024-03878-x](https://doi.org/10.1007/s00348-024-03878-x). arXiv: [2403.11828](https://arxiv.org/abs/2403.11828).
- [31] L. Schena, P. Marques, R. Poletti, S. Ahizi, J. Van den Berghe, and M. A. Mendez. “Reinforcement Twinning: from digital twins to model-based reinforcement learning”. In: *Journal of Computational Science* 82 (2024), p. 102421. DOI: [10.1016/j.jocs.2024.102421](https://doi.org/10.1016/j.jocs.2024.102421). arXiv: [2311.03628](https://arxiv.org/abs/2311.03628).
- [32] P. Marques, S. Ahizi, and M. A. Mendez. “Real-time data assimilation for the thermodynamic modelling of cryogenic storage tanks”. In: *Energy* 302 (2024), p. 131739. DOI: [10.1016/j.energy.2024.131739](https://doi.org/10.1016/j.energy.2024.131739). arXiv: [2310.11399](https://arxiv.org/abs/2310.11399).
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- [34] P. Sperotto, M. Ratz, and M. A. Mendez. “SPICY: a Python toolbox for meshless assimilation from image velocimetry using radial basis functions”. In: *Journal of Open Source Software* 9.93 (2024), p. 5749. DOI: [10.21105/joss.05749](https://doi.org/10.21105/joss.05749).
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- [41] F. Pino, L. Schena, J. Rabault, and M. A. Mendez. “Comparative analysis of machine learning methods for active flow control”. In: *Journal of Fluid Mechanics* 958 (2023), A39. DOI: [10.1017/jfm.2023.76](https://doi.org/10.1017/jfm.2023.76). arXiv: [2202.11664](https://arxiv.org/abs/2202.11664).
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